Introduction to Photosynthesis

- Photosynthesis is the primary process by which plants, algae, and some bacteria convert light energy into chemical energy [S1].
- This biological process occurs within specialized organelles called chloroplasts found in plant cells [S1].
- The overall reaction transforms carbon dioxide and water into glucose and oxygen using sunlight.
- It provides the foundational energy source for nearly all life forms on Earth.
- The process is divided into two main stages: the light-dependent reactions and the light-independent Calvin cycle [S1].



Sunbeams Filtering Through the Forest Canopy at National Botanical Garden, Bangladesh — Mahbub Hasan Mreedul (CC BY-SA 4.0), Wikimedia Commons

Agenda

- Overview of chloroplast structure and the site of reactions.
- Detailed breakdown of the light-dependent reactions in the thylakoid.
- Analysis of the light-independent reactions, also known as the Calvin cycle.
- The role of key molecules like ATP, NADPH, and the enzyme RuBisCO [S1].
- A worked example of the chemical equation and energy balance.
- Summary of key concepts and a concluding discussion.

Learning Objectives

- Identify the structural components of the chloroplast and their specific functions [S1].
- Describe the mechanism of water splitting and oxygen release in the thylakoid membranes [S1].
- Explain the production and role of ATP and NADPH as energy carriers [S1].
- Outline the steps of carbon fixation within the stroma via the Calvin cycle [S1].
- Understand the critical role of the enzyme RuBisCO in glucose production [S1].

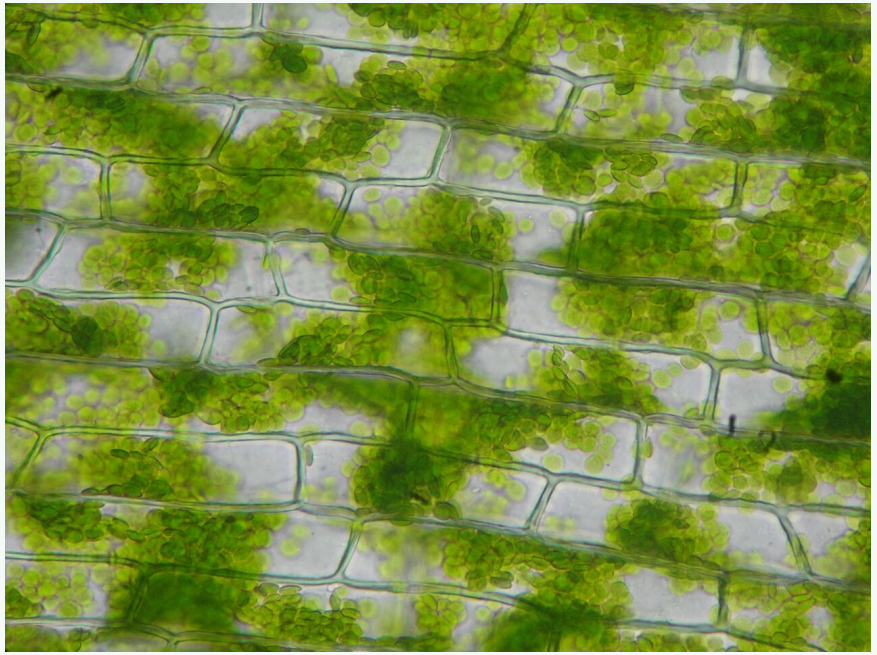
The Chloroplast: The Site of Photosynthesis

- Chloroplasts are double-membrane organelles that house the machinery for photosynthesis [S1].
- The thylakoids are flattened sac-like membranes arranged in stacks called grana.
- The stroma is the fluid-filled space surrounding the thylakoids where the Calvin cycle occurs [S1].
- Chlorophyll pigments embedded in the thylakoid membranes capture light energy.
- The compartmentalization of the chloroplast allows different chemical environments for different reactions.



Light-Dependent Reactions

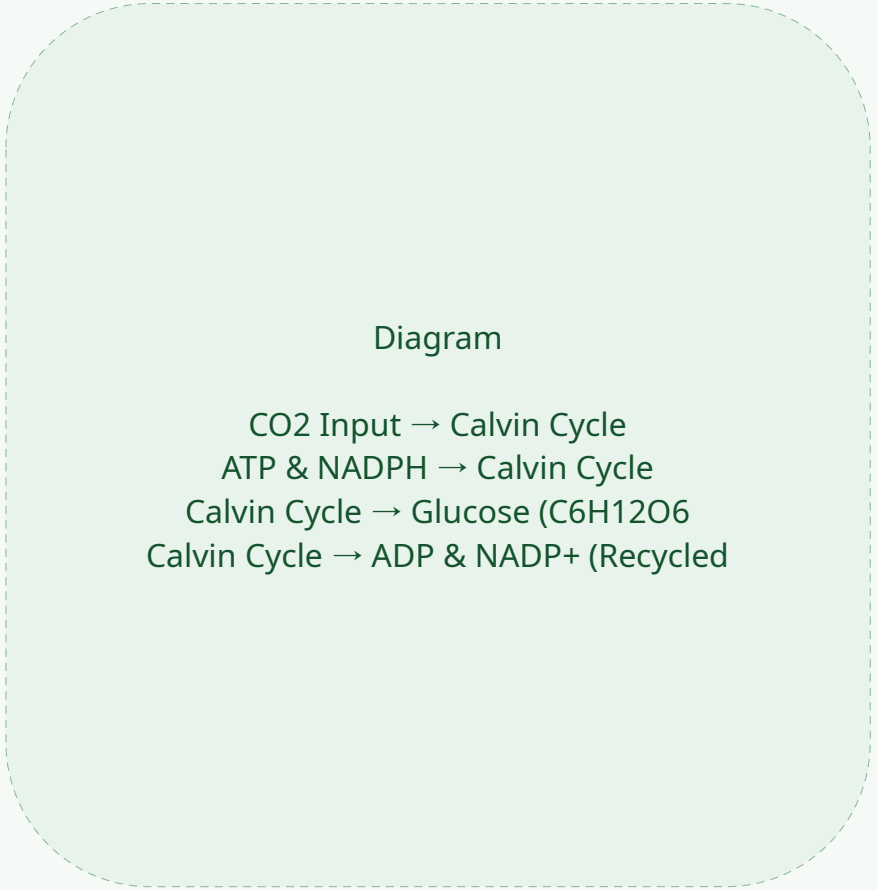
- These reactions take place specifically within the thylakoid membranes of the chloroplast [S1].
- Light energy is used to split water molecules (photolysis), which releases oxygen as a byproduct [S1].
- The process generates high-energy electrons that move through an electron transport chain.
- This energy is captured to produce two vital molecules: ATP (adenosine triphosphate) and NADPH [S1].
- These two molecules act as "energy currency" for the next stage of photosynthesis.



Chloroplasts - Microscopic view of Elodea canadensis — Juan Carlos Fonseca Mata (CC BY-SA 4.0), Wikimedia Commons

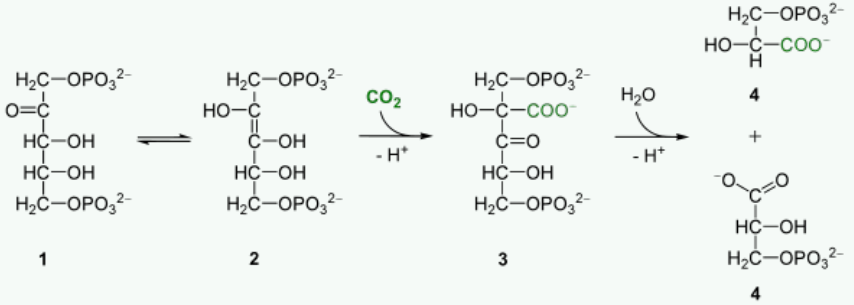
The Calvin Cycle: Carbon Fixation

- The Calvin cycle occurs in the stroma and does not directly require light [S1].
- It utilizes the ATP and NADPH produced during the light-dependent reactions [S1].
- Carbon dioxide (CO_2) is taken from the atmosphere and "fixed" into organic molecules [S1].
- The primary end product of this cycle is glucose, a simple sugar used for energy and growth.
- This cycle must turn multiple times to produce a single molecule of glucose.



The Role of RuBisCO

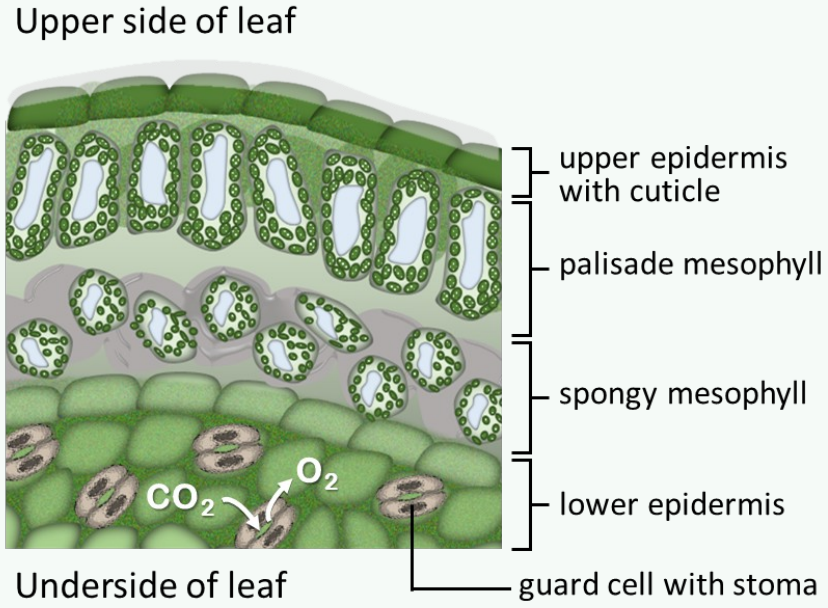
- RuBisCO (Ribulose-1,5-bisphosphate carboxylase/oxygenase) is the enzyme that catalyzes the first step of the Calvin cycle [S1].
- It is responsible for attaching CO_2 to a five-carbon sugar called RuBP [S1].
- RuBisCO is often cited as the most abundant protein on Earth because it is so essential for life.
- Without this enzyme, the conversion of inorganic carbon to organic matter would be too slow to support life.
- It is a highly regulated enzyme that is sensitive to temperature and gas concentrations.



Rubisco com carboxilasa — Yikrazuul (CC BY-SA 4.0), Wikimedia Commons

Worked Example: The Photosynthesis Equation

- The balanced chemical equation for photosynthesis is: $6CO_2 + 6H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6O_2$.
- Six molecules of carbon dioxide and six molecules of water are required to produce one molecule of glucose.
- The oxygen released ($6O_2$) comes entirely from the water molecules, not the carbon dioxide.
- This equation represents a redox process where CO_2 is reduced and H_2O is oxidized.
- Energy is stored in the chemical bonds of the glucose molecule ($C_6H_{12}O_6$).



Cross section of leaf — ELaurent (CC BY-SA 4.0), Wikimedia Commons

Recap of Key Concepts

- Photosynthesis happens in the chloroplast: light reactions in thylakoids, Calvin cycle in stroma [S1].
- Light reactions: Split H_2O, release O_2, and create ATP/NADPH [S1].
- Calvin cycle: Use ATP/NADPH to fix CO_2 into glucose using RuBisCO [S1].
- Energy transformation: Light energy is converted into stable chemical energy in sugars.
- Biological importance: This process supports the food chain and regulates atmospheric oxygen.

Discussion and Assessment

- How would a lack of water affect the production of ATP and NADPH?
- Why is the Calvin cycle often called the "dark reactions" even though it usually happens during the day?
- Predict the impact on a plant if the enzyme RuBisCO were to become denatured by high heat.
- Discuss the relationship between the light-dependent and light-independent reactions—can one happen without the other?
- Review: What are the three main inputs and two main outputs of the overall process?